

$$\therefore f = (a+d) \cdot (a+\bar{b}+\bar{c}) \cdot (\bar{a}+b+d)$$

Day - 03

Combinational Assignment

1) Implement the following functions using

i) 2:1 mux and not gates

ii) 4:1 mux

a) $F(A, B, C) = \bar{A}C + B\bar{C}$

b) $F(P, Q, R) = \bar{P}\bar{R} + Q\bar{R} + \bar{P}Q$

Soln

a) $F(A, B, C) = \bar{A}C + B\bar{C}$

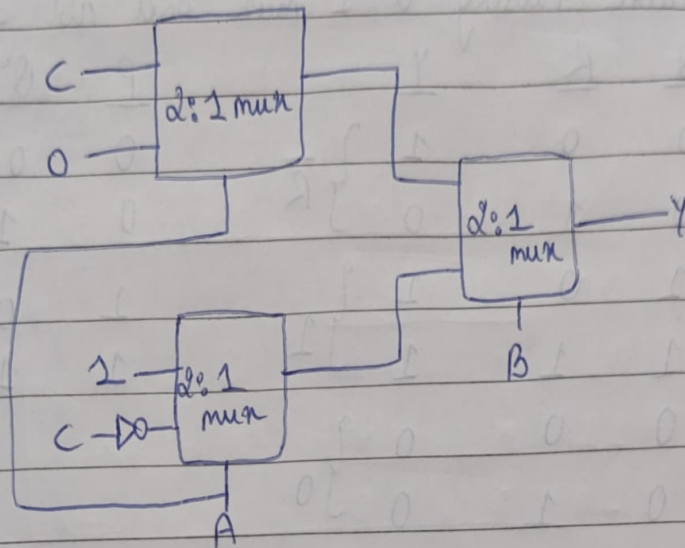
i) $= \bar{A}C(B+\bar{B}) + (A+\bar{A})B\bar{C}$

$= \bar{A}BC + \bar{A}\bar{B}C + ABC + \bar{A}B\bar{C}$

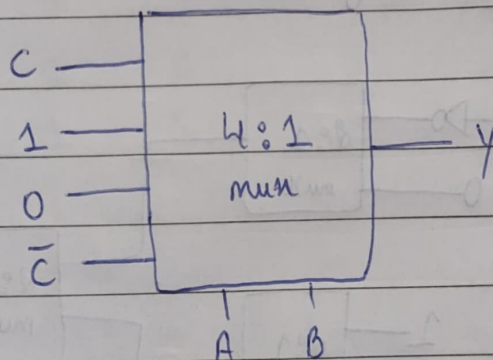
$= m(1, 2, 3, 6)$

Truth Table (using 2:1 mux and not gates)

<u>A</u>	<u>B</u>	<u>C</u>	<u>Y(output)</u>		<u>A</u>	<u>B</u>	<u>Y</u>
0	0	0	0	} C	0	0	C
0	0	1	1		0	1	1
0	1	0	1	} 1	1	0	0
0	1	1	1		1	1	$\bar{C}$
1	0	0	0	} 0			
1	0	1	0				
1	1	0	1	} $\bar{C}$			
1	1	1	0				



ii) using 4:1 mux (using the previous truth table)



$$b) F(P, Q, R) = \bar{P}\bar{R} + Q\bar{R} + \bar{P}Q$$

$$i) Y = \bar{P}\bar{R}(Q + \bar{Q}) + (P + \bar{P})Q\bar{R} + \bar{P}Q(R + \bar{R})$$

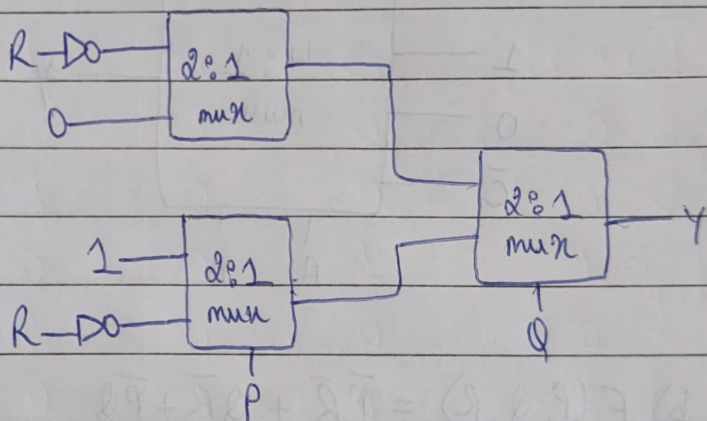
$$Y = \bar{P}\bar{R}Q + \bar{P}\bar{R}\bar{Q} + P\bar{R}Q + \bar{P}\bar{R}\bar{Q} + P\bar{R}Q + \bar{P}\bar{R}\bar{Q} + P\bar{R}Q + \bar{P}\bar{R}\bar{Q}$$

$$Y = \bar{P}\bar{R}Q + \bar{P}\bar{R}\bar{Q} + P\bar{R}Q + \bar{P}\bar{R}\bar{Q}$$

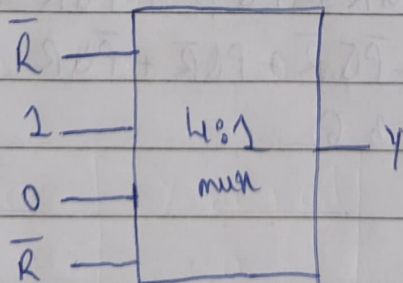
$$Y = m(0, 2, 3, 6)$$

Truth Table (using 2:1 mux and not gate)

<u>P</u>	<u>Q</u>	<u>R</u>	<u>Y</u>		<u>P</u>	<u>Q</u>	<u>Y</u>
0	0	0	1	$\left. \begin{array}{l} \overline{R} \\ R \end{array} \right\}$	0	0	$\overline{R}$
0	0	1	0		0	1	1
0	1	0	1	$\left. \begin{array}{l} 1 \\ 1 \end{array} \right\}$	1	0	0
0	1	1	1		1	1	$\overline{R}$
1	0	0	0	$\left. \begin{array}{l} 0 \\ 0 \end{array} \right\}$			
1	0	1	0				
1	1	0	1	$\left. \begin{array}{l} \overline{R} \\ R \end{array} \right\}$			
1	1	1	0				



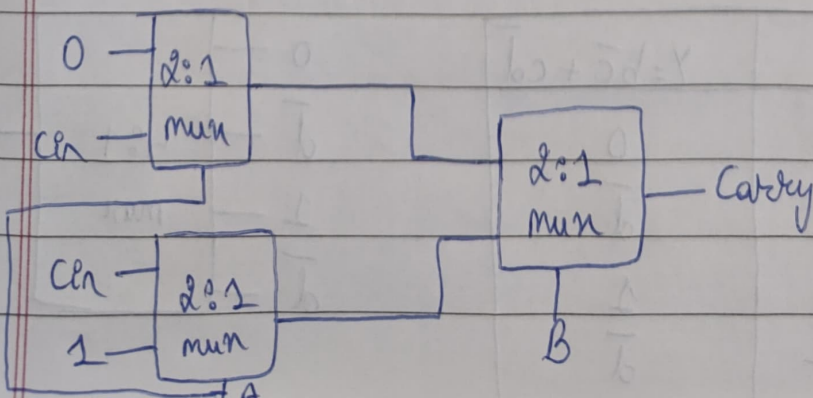
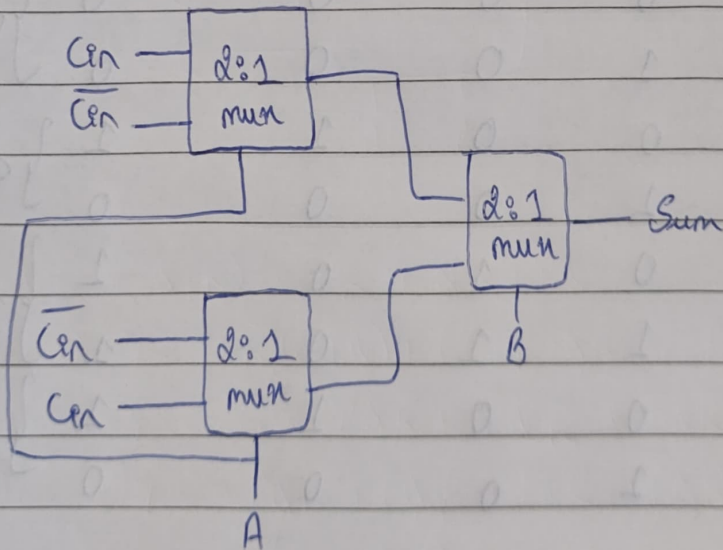
ii) using 4:1 mux



2. Implement the functionality of a full adder using 2:1 Muxes only

Solⁿ Truth Table of Full Adder

A	B	Cin	Sum	Carry		A	B	Sum	Carry
0	0	0	0	0		0	0	Cin	0
0	0	1	1	0		0	1	$\bar{C}_{in}$	Cin
0	1	0	1	0	$\Rightarrow$	1	0	$\bar{C}_{in}$	Cin
0	1	1	0	1		1	1	Cin	1
1	0	0	1	0					
1	0	1	0	1					
1	1	0	0	1					
1	1	1	1	1					



3. Simplify $f(a, b, c, d) = \sum m(2, 4, 5, 6, 10) + \sum d(12, 13, 14, 15)$ and implement SOP expression using 4:1 mux

Soln

ab \ cd	00	01	11	10
00	0	0	0	1
01	1	1	0	1
11	X	X	X	X
10	0	0	0	1

$$y = b\bar{c} + c\bar{d}$$

Let b & c be the select lines

b	c	d	$b\bar{c}$	$c\bar{d}$	$b\bar{c} + c\bar{d}$
0	0	0	0	0	0
0	0	1	0	0	0
0	1	0	0	1	1
0	1	1	0	0	0
1	0	0	1	0	1
1	0	1	1	0	1
1	1	0	0	1	1
1	1	1	0	0	0

b	c	$Y = b\bar{c} + c\bar{d}$	
0	0	0	0 —
0	1	$\bar{d}$	$\bar{d}$ —
1	0	$\bar{d}$	1 —
1	1	$\bar{d}$	$\bar{d}$ —

4:1 mux — Y